

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Mallavarapu V Venkata Satyanarayana Swamy	Reg. No.:	192425063
Section / Batch:	CSE-AI/2024	Date:	27-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 – 50 Q2 – 50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Consider the concept map: Algorithm → Growth Function → Big-Omega → Lower Bound → Best-case Explain how lower bound represents minimum execution time. Extend the map with performance guarantee node. Justify why best-case analysis is useful. Interpret how this contributes to overall performance understanding.

Code of Conduct

I Mallavarapu V Venkata Satyanarayana Swamy(192425063) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.Swamy

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Lower Bound and Performance Understanding

1. How Lower Bound Represents Minimum Execution Time.

The Lower Bound of an algorithm is the minimum amount of time Required to Solve a Problem, regardless of the algorithm used. It represents the theoretical limit below which no algorithm can perform for that Problem.

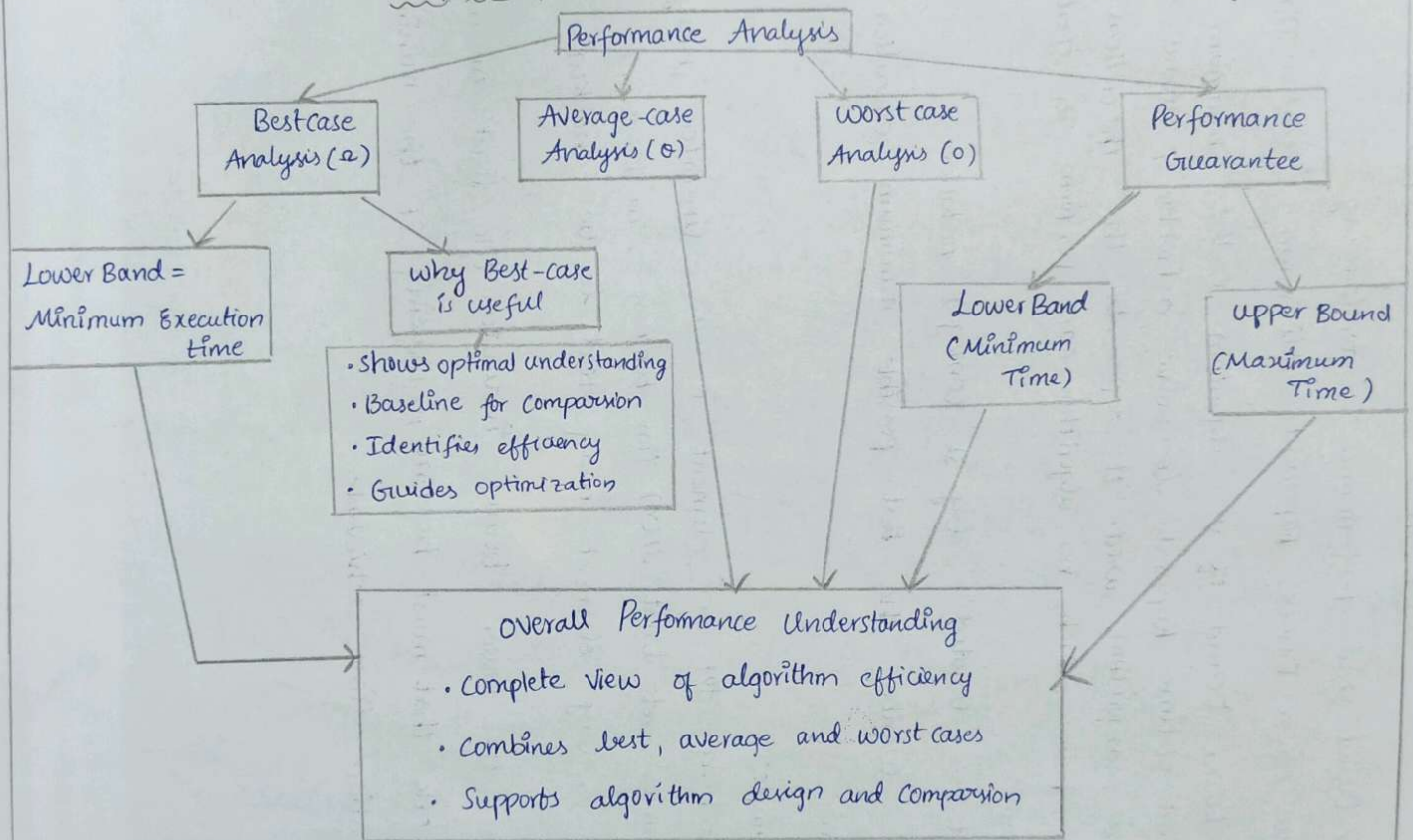
- It is denoted by Ω (omega) notation.
- It describes the best possible minimum execution time

• Example:

Searching for an element in an unsorted array requires at least $\Omega(n)$ time because, in the worst case, every element may need to be checked.

Thus, the lower bound indicates the least amount of work that must be done, making it the minimum execution time achievable.

Concept Map: Lower Bound and Performance Understanding



why Best-case Analysis is useful

Best-case analysis is useful because:

- It shows the minimum execution time an algorithm can achieve.
- It helps identify the algorithm's optimal performance.
- It is useful when inputs are already sorted or favorable.
- It provides a baseline for comparison with average and worst-case performance.
- It helps evaluate whether further optimization is possible.

Contribution to overall Performance Understanding.

Lower bound and best-case analysis improve performance understanding by:

- Identifying the minimum possible running time of a problem
- Showing the algorithm's optimal efficiency
- Helping compare theoretical and practical performance.
- Supporting the design of more efficient algorithms
- Providing a complete view when combined with average-case and worst-case analysis.

conclusion:

The lower bound (Ω) represents the Minimum execution time required to solve a problem and serves as a theoretical performance limit. Extending the performance map with a performance Guarantee node connects lower and upper bounds, giving a complete picture of algorithm efficiency. Best-case analysis is valuable because it highlights optimal performance and contributes to a deeper understanding of overall algorithm behavior.